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EE25BTECH11002 - Achat Parth Kalpesh

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Question

In a linear inverse problem, the coefficient matrix, $\mathbf{A} = \begin{pmatrix} 2.00 & 2.01 \\ 2.01 & 2.00 \end{pmatrix}$.

The eigenvalues of $\mathbf{A}$ are

1. $(4.01\,, -0.01)$
2. $(-4.01\,, -0.01)$
3. $(4.01\,, 0.01)$
4. $(-4.01\,, 1.01)$

Solution:

The characteristic equation for matrix $\mathbf{A}$ is given by

$$\left|\mathbf{A} - \lambda\mathbf{I}\right| = 0 \tag{1}$$

$$\begin{vmatrix} 2.00 - \lambda & 2.01 \\ 2.01 & 2.00 - \lambda \end{vmatrix} = 0 \tag{2}$$

$$(2.00 - \lambda)^2 - (2.01)^2 = 0 \tag{3}$$

$$((2.00 - \lambda) - 2.01)\left((2.00 - \lambda) + 2.01\right) = 0 \tag{4}$$

$$(-\lambda - 0.01)(-\lambda + 4.01) = 0 \tag{5}$$

$$\therefore \lambda_1 = 4.01, \quad \lambda_2 = -0.01 \tag{6}$$

$$\tag{7}$$

Solution:

Hence, the eigenvalues are

$$\lambda_1 = 4.01,\ \lambda_2 = -0.01 \tag{8}$$